GEOGRAPHIC LOCALIZATION OF KNOWLEDGE SPILLOVERS AS EVIDENCED BY PATENT CITATIONS*

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We compare the geographic location of patent citations with that of the cited patents, as evidence of the extent to which knowledge spillovers are geographically localized. We find that citations to domestic patents are more likely to be domestic, and more likely to come from the same state and SMSA as the cited patents, compared with a "control frequency" reflecting the pre-existing concentration of related research activity. These effects are particularly significant at the local (SMSA) level. Localization fades over time, but only very slowly. There is no evidence that more "basic" inventions diffuse more rapidly than others.

The last decade has seen the development of a significant body of empirical research on R&D spillovers.[1] Generally speaking, this research has shown that the productivity of firms or industries is related to their R&D spending, and also to the R&D spending of other firms or other industries. In parallel, economic growth theorists have focused new attention on the role of knowledge capital in aggregate economic growth, with a prominent modeling role for knowledge spillovers (e.g., Romer [1986, 1990] and Grossman and Helpman [1991]).

We know very little, however, about where spillovers go. Is there any advantage to nearby firms, or even firms in the same country, or do spillovers waft into the ether, available for anyone around the globe to grab? The presumption that U. S. international competitiveness is affected by what goes on at federal laboratories and U. S. universities, and the belief that universities and other research centers can stimulate regional economic growth[2] are predicated on the existence of a geographic component to the

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1. E.g., Jaffe [1986] and Bernstein and Nadiri [1988, 1989]. For a recent survey and evaluation of this literature, see Griliches [1991].

2. See, e.g., Minnesota Department of Trade and Economic Development [1988]; Dorfman [1988]; Feller [1989]; and Smilor, Kozmetsky, and Gibson [1988].

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spillover mechanism. The existing spillover literature, however, is virtually silent on this point.[3]

In the growth literature it is typically assumed that knowledge spills over to other agents within the country, but not to other countries.[4] This implicit assumption begs the question of whether and to what extent knowledge externalities are localized. As emphasized recently by Krugman [1991], acknowledging the importance of spillovers and increasing returns requires renewed attention by economists to issues of economic geography. Krugman revives and explores the explanations given by Marshall [1920] as to why industries are concentrated in cities. Marshall identified three factors favoring geographic concentration of industries: (1) the pooling of demands for specialized labor; (2) the development of specialized intermediate goods industries; and (3) knowledge spillovers among the firms in an industry. Krugman believes that economists should focus on the first two of these, partially because he perceives that "[k]nowledge flows, by contrast, are invisible; they leave no paper trail by which they may be measured and tracked, and there is nothing to prevent the theorist from assuming anything about them that she likes" [Krugman, p. 53].

But knowledge flows do sometimes leave a paper trail, in the form of citations in patents. Because patents contain detailed geographic information about their inventors, we can examine where these trails actually lead. Subject to caveats discussed below relative to the relationship between citations and spillovers, this allows us to use citation patterns to test the extent of spillover localization. We examine citations received by patents assigned to universities, and also the citations received by a sample of domestic corporate patents. If knowledge spillovers are localized within countries, then citations of patents generated within the United States should come disproportionately from within the United States. To the extent that regional localization of spillovers is

3. Jaffe [1989] provides evidence that corporate patenting at the state level depends on university research spending, after controlling for corporate R&D. Mansfield [1991] surveyed industrial R&D employees about university research from which they benefited. He found that they most often identified major research universities, but that there was some tendency to cite local universities even if they were not the best in their field.

4. The existence of this implicit assumption was noted by Glaeser, Kallal, Scheinkman, and Shleifer [1991]: "After all, intellectual breakthroughs must cross hallways and streets more easily than oceans and continents." Grossman and Helpman [1991] consider international knowledge spillovers explicitly.

important, citations should come disproportionately from the same state or metropolitan area as the originating patent.[5]

The most difficult problem confronted by the effort to test for spillover-localization is the difficulty of separating spillovers from correlations that may be due to a pre-existing pattern of geographic concentration of technologically related activities. That is, if a large fraction of citations to Stanford patents comes from the Silicon valley, we would like to attribute this to localization of spillovers. A slightly different interpretation is that a lot of Stanford patents relate to semiconductors, and a disproportionate fraction of the people interested in semiconductors happen to be in the Silicon valley, suggesting that we would observe localization of citations even if proximity offers no advantage in receiving spillovers. Of course, the ability to receive spillovers is probably one reason for this pre-existing concentration of activity. If it were the *only* possible reason, then, under the null hypothesis of no spillover localization we should still see no localization of citations. As discussed above, however, there are other sources of agglomeration effects that could explain the geographic concentration of technologically related activities without resort to localization of knowledge spillovers. For this reason, we construct "control" samples of patents that are not citations but have the same temporal and technological distribution as the citations. We then calculate the geographic matching frequencies between the citations and originating patents, and between the controls and originating patents. Our test of localization is whether the citation matching frequency is significantly greater than the control matching frequency. Since the "control" matching frequency is, itself, likely to be partly the result of spillover-localization, we believe this to be a conservative test for the existence of localization.

The first section of the paper describes patents and patent citations, explains the construction of the control samples, and

5. Glaeser, Kallal, Scheinkman, and Shleifer [1991] characterize the "Marshall-Arrow-Romer" models as focusing on knowledge spillovers within the firms in a given industry. They examine the growth rate of industries in cities as a function of the concentration of industrial activity across cities, within-city industrial diversity, and within-city competition. They find that within-city diversity is positively associated with growth of industries in that city, while concentration of an industry within a city does not foster its growth. They interpret this contrast to mean that spillovers across industries are more important than spillovers within industries. As is discussed below, there is evidence from the R&D spillover literature to suggest that across-industry knowledge spillovers are, indeed, important. In this study, we do not consider the *industrial* identity of either generators or receivers of spillovers, though we do have some information on their *technological* similarity.

considers more carefully how citations might be used to infer spillovers. The second section presents the results of the tests of geographic localization. The following section examines whether the probability of geographic localization of any given citation can be explained by attributes of the originating or citing patents, or of relationships between them. A concluding section follows.

I. EXPERIMENTAL DESIGN

A. Patents and Patent Citations[6]

A patent is a property right in the commercial use of a device.[7] For a patent to be granted, the invention must be nontrivial, meaning that it would not appear obvious to a skilled practitioner of the relevant technology, and it must be useful, meaning that it has potential commercial value. If a patent is granted,[8] a public document is created containing extensive information about the inventor, her employer, and the technological antecedents of the invention, all of which can be accessed in computerized form. Among this information are "references" or "citations." It is the patent examiner who determines what citations a patent must include. The citations serve the legal function of delimiting the scope of the property right conveyed by the patent. The granting of the patent is a legal statement that the idea embodied in the patent represents a novel and useful contribution over and above the previous state of knowledge, as represented by the citations. Thus, in principle, a citation of Patent X by Patent Y means that X represents a piece of previously existing knowledge upon which Y builds.

The examiner has several means of identifying potential citations. The applicant has a legal duty to disclose any knowledge of the prior art that she may have. In addition, the examiner is supposed to be an expert in the technological area and be able to

6. All of the data we use relate to patents granted by the U. S. patent office. About 40 percent of U. S. patents are currently granted to foreigners. Other countries, of course, also grant patents, leading to some ambiguity in the meaning of phrases like "U. S. patent" and "foreign patent." We shall use the phrase "U. S. patent" to mean a patent granted by the U. S. patent office, regardless of the residence of the inventor. We shall use the phrase "domestic patent" to refer to a patent granted (by the U. S. patent office) to an inventor residing in the United States. We shall use the phrase "foreign patent" to refer to a patent granted by the U. S. patent office to non-U. S. residents.

7. Ideas are not patentable; nor are algorithms or computer programs, though a chip with a particular program coded into it might be. The definition of a device was recently broadened to include genetically engineered organisms.

8. There is no public record of unsuccessful patent applications.

identify relevant prior art that the applicant misses or conceals. The framework for the search of the prior art is the patent classification system. Every patent is assigned to a nine-digit patent class (of which there are about 100,000) as well as an unlimited number of additional or "cross-referenced" classes. An examiner will typically begin the search of prior art using her knowledge of the relevant classes. For the purpose of identifying distinct technical areas, we utilize aggregations of subclasses to a three-digit level; at this level there are currently about 400 technical classes.[9]

For this study, we begin with two cohorts of "originating" patents, one consisting of 1975 patent applications and the other of 1980 applications. In each cohort we include all patents granted to U. S. universities and two samples of U. S. corporate patents[10] chosen to match the university patents by grant date and technological distribution. These sets of originating patents were chosen because we conjectured that the extent of geographic localization might differ depending on the nature of the originating institution. As discussed below, such differences turn out to be minor. The 1975 originating cohort contains about 950 patents that had received a total of about 4750 citations by the end of 1989. The 1980 originating cohort contains about 1450 that had received about 5200 citations by the same time.

B. Construction of "Control" Samples

The main idea of this paper is to compare the geographic location of the citations with the originating patent that they cite. But to make such a comparison meaningful, we have to consider how often we would expect them to match under some "null" hypothesis. That is, we need to compare the probability of a patent matching the originating patent by geographic area, *conditional* on its citing the originating patent, with the probability of a match *not conditioned on the existence of a citation link.* This noncitation-conditioned probability gives us a baseline or reference value against which to compare the proportions of citations that match.

9. Examples of three-digit patent classes are "Batteries, Thermoelectric and Photoelectric"; "Distillation: Apparatus"; "Robots"; seventeen distinct classes of "Organic Compounds"; and the ever-popular "Whips and Whip Apparatus."

10. The "top corporate" sample consists of patents granted to the 200 top-R&D-performing firms in the United States, as reported in S.E.C. 10-k forms and compiled by Compustat. The "other corporate" sample contains patents assigned to U. S. corporations that are not universities and not in the "top corporate" sample.

We call this baseline or reference probability the "control frequency."

Two considerations drove our choice for constructing the control frequency. First, the fraction of U. S. patents granted to foreigners has been climbing steadily during the period under study here. We do not want to conclude that citations are initially localized, but that this localization fades over time, simply because of this aggregate trend. Second, countries (and cities and states) differ in their areas of technological focus. Although such technological specialization is probably due, in part, to geographic localization of spillovers, we want to be conservative and test whether spillovers are localized *relative to what would be expected given the existing distribution of technological activity.*

To derive a control frequency that would be immune to contamination from either aggregate movements over time or localization based on the pre-existing concentration of technological activity, we went back to the patent data base and found a "control patent" to correspond to each of the citing patents. For each citing patent, we identified all patents in the *same patent class* with the *same application year* (excluding any other patents that cited the same originating patent). We then chose from that set a control patent whose grant date was as close as possible to that of the citing patent. This process yielded, for each set of citing patents, a corresponding control sample of equal size, whose distribution across time and technological areas is essentially identical to that of the citation data set. Each control patent is paired with a particular citing patent, allowing us to compare the geographic location of the control patent with that of the originating patent cited by its counterpart in the citing dataset. The frequency with which these control patents match geographically with the originating patent is an estimate of the frequency with which a randomly drawn patent that is not a citation, but has the same technological and temporal profile as the citation, matches geographically.

To put it slightly differently, when we calculate the frequency with which the citations match the geographic location of the cited patents, we are estimating the probability of geographic match for two patents, *conditional on there being a citation link and also conditional on the technological nature and timing of the citation.* When we calculate the frequency with which the "control" patents match geographically with the cited patents, we are estimating the probability of geographic match for two patents, *conditional only*

hypothetical contract the assumption that such contracted development is relatively likely to be localized, we have the potential for the observed localization of citations to be greater than the true localization of knowledge spillovers.[11]

Although such "internalized spillovers" surely exist, it is likely that most citations that are not spillovers are of a different sort: citations (added by the examiner) to previous patents of which the citing inventor was unaware. Clearly, no spillover occurs in this case. Further, it seems likely that citations of this sort should not be any more geographically localized than the control patents. If many citations are in this category, it introduces "noise" into the citations as a measure of spillovers, and biases the results *away* from finding significant localization. Our a priori belief is that this category is much larger than the previous one, suggesting that spillovers are, on balance, probably more localized than citations, but readers with different beliefs should interpret our results accordingly.

Finally, there are an enormous number of spillovers with no citations, since only a small fraction of research output is ever patented. In particular, much of the results of very basic research cannot be patented. It is plausible that basic research generates the largest spillovers,[12] and also that basic research is communicated via mechanisms that are less likely to be localized, such as international journals. For this reason, it is probably appropriate to view our results as related to applied research, and to exercise care in extrapolating to the localization of spillovers from extremely basic research.

D. Geographic Assignment of Patents

The preceding discussion has presumed that the "location" of a patent is an unambiguous construct. The patent data contain the country of residence of each inventor, and the city and state of

11. As discussed below, we focus on tests of localization that exclude citing patents that are owned by the same organization as the originating patent, precisely because such "self-citations" do not represent an externality. Citations by other organizations that have an economic relationship with the original inventor could be viewed as similar to self-citations. There is, however, a significant difference: we expect that, in general, the contract between the two parties will be quite incomplete, making it more likely than not that the citing organization could capture some rents from the original invention and hence benefit from at least a partial spillover.

12. This question is analyzed in detail in Trajtenberg, Henderson, and Jaffe [1992].

residence for U. S. inventors.[13] Use of this information is complicated by the fact that patents can have multiple inventors who can live in different places. The following procedure was followed:

1. For U. S. *inventors,* city/state combinations were placed in counties using a commercially available city directory; each U. S. inventor was then assigned to an SMSA[14] based on state and county. For this purpose an additional "phantom" SMSA was created in each state, encompassing all counties in the state outside of defined SMSAs. Approximately 98 percent of inventors were successfully assigned to SMSAs.

2. Assignments of each *patent* to a country, a state, and an SMSA were then made based on pluralities of inventors. So, for example, a patent with one inventor living in Bethesda, MD, one in Alexandria, VA, and one in rural Virginia would be assigned VA for its state and Washington, DC, for its SMSA. Ties were assigned arbitrarily, except that ties between true SMSAs and phantom SMSAs were resolved for the true one and ties between United States and foreign were resolved in favor of foreign.[15]

II. Results on Extent of Localization

As a prelude to the geographic analysis, Table I and Figure I present some descriptive data about the citations and their relationship to the originating patents. Table I shows that about 80–90 percent of the 1975 patents and 70–80 percent of the 1980 patents had received at least one citation by the end of 1989, with the higher proportion in each case applying to the university patents. Mean citations received (including zeros) were four–six for 1975 and three–four for 1980, again with the higher numbers corresponding to the university patents.[16] The average lag between the originating application year and the application year of the citing

13. Published data on the geographic distribution of U. S. patents (including those cited in the popular press) are based on the location of the organization or individual to which the patent is "assigned" by the inventor (usually her employer). For analysis of localization, the location of the assignee is not a desirable datum. There is an ambiguity in such data relating to the way employees of multinational corporations make their assignments. An employee of "Honda" in the United States could assign her patent to "Honda U.S.A., Inc." or she could assign it to the parent company in Japan. In the former case, the patent office would call it a domestic patent, and in the latter case a foreign patent; similarly for IBM Switzerland.

14. These assignments were made based on the 1981 SMSA definitions. In areas where Consolidated Metropolitan Statistical Areas were defined in 1981, these were used; elsewhere Metropolitan Statistical Areas were used. Hence we use the generic term "SMSA."

15. At the country level, 98 percent of patents were assigned unanimously. At the state level, 90 percent were assigned unanimously; an additional 4 percent had more than half of inventors in a single state. At the SMSA level, 86 percent were assigned unanimously, and an additional 6 percent had a clear majority. Overall, 4.5 percent of patents were assigned to "phantom" SMSAs.

16. Our companion paper [Trajtenberg, Henderson, and Jaffe, 1992] explores in detail the use of citation intensity and related measures for measuring the basicness and appropriability of inventions.

TABLE I
DESCRIPTIVE STATISTICS

Originating dataset	Percent receiving citations	Total no. of citations	Mean citations received	Average citation lag[a,b]	Percent self-citations[b]	Percent same patent class[b,c]
1975						
University	88.6	1933	6.12	6.53	5.6	54.3
Top corporate	84.2	1476	4.70	7.17	18.6	55.7
Other corporate	82.3	1341	4.22	7.82	9.1	57.5
1980						
University	79.9	2093	4.34	4.36	8.9	56.3
Top corporate	79.9	1701	3.54	4.41	24.6	58.3
Other corporate	74.1	1424	2.95	4.46	12.6	57.2

a. Application year of citing patent minus application year of originating patent.
b. For those patents receiving any citations.
c. Comparison is at the three-digit level (see text).

patent is 6.5 to 8 years for the 1975 cohort, and a little over 4 years for the 1980 cohort.

The inference that a citation indicates a possible knowledge spillover is much less clear in the case where the citing patent is owned by the same organization as the originating patent. For this reason, we distinguish what we call "self-citations." A self-citation is defined as a citing patent assigned by its inventors to the same party as the originating patent, which is, by construction, either a university or a domestic corporation. Not surprisingly, the self-citation rate differs for the different sources of originating patents, with universities having the lowest and top corporations the highest rates.[17] Finally, Table I shows that 55 to 60 percent of citations have a primary patent class that is the same as the primary patent class of the originating patent, indicating that the originating and citing patents are *technologically* close to one another.

Figure I provides additional detail on the distribution of lags between originating and citing patents, again defined as the difference in application years. The figure shows that citations are few in the early years,[18] and reach a plateau after about three years.

17. The apparent increase in self-citation rates between 1975 and 1980 is probably spurious; self-citations tend to come earlier than other citations. See Trajtenberg, Henderson, and Jaffe [1992] for more on this issue.

18. Patents are typically granted one to three years after application; thus, a citation lag of zero or one implies that the citing patent may well have been applied for before the originating patent was actually granted. Pending applications are not public, so in this case the citation would almost surely have been identified by the examiner.

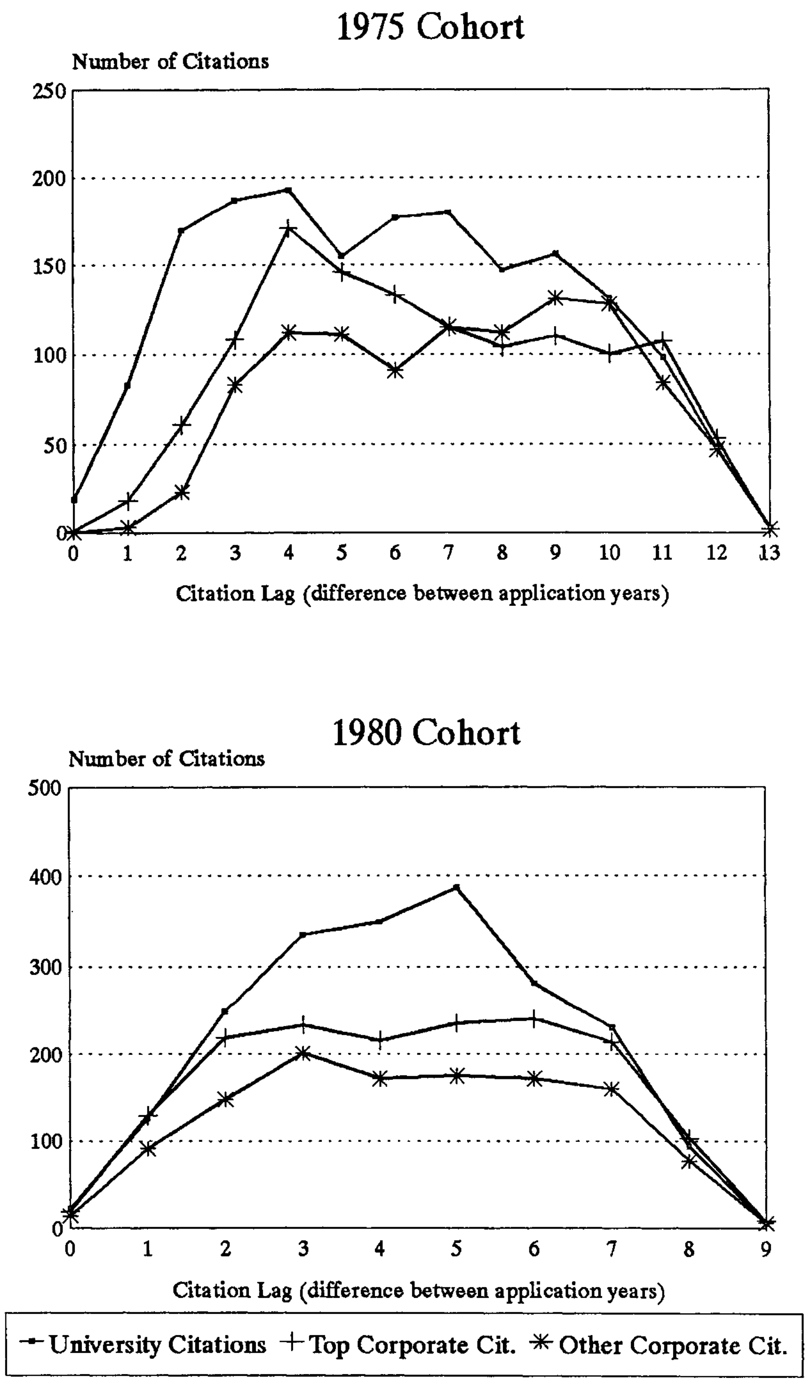
1975 Cohort
Number of Citations
250
200
150
100
50
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Citation Lag (difference between application years)
1980 Cohort
Number of Citations
500
400
300
200
100
0
0 1 2 3 4 5 6 7 8 9
Citation Lag (difference between application years)
University Citations
Top Corporate Cit.
Other Corporate Cit.

FIGURE I

TABLE II
SMSA Distributions for Some Datasets

Location	1975 University originating	1975 Top corporate originating	Citations to 1975 university	Citations to 1975 top corporate	Controls for citations to 1975 university	All citations to patents from in NY SMSA
Foreign	—	—	31.8	31.4	35.8	31.2
Boston	15.0	3.1	7.5	4.6	5.1	4.0
Los Angeles/ Anaheim	7.0	4.8	9.0	5.7	6.1	3.9
San Francisco/ Oakland	5.1	1.4	3.8	3.7	6.1	3.5
Madison, WI	4.2	—	1.6	—	0.5	0.6
Philadelphia/ Wilmington	4.2	9.3	5.4	8.2	4.5	9.1
Rural Iowa	3.8	—	1.6	0.6	0.2	—
San Jose	3.5	2.8	4.0	3.4	—	1.9
New York/ NJ/CT	3.2	13.5	9.7	11.7	13.7	28.5
Salt Lake City	3.2	—	2.1	—	0.5	0.4
Detroit/ Ann Arbor	2.6	2.4	2.6	1.7	1.7	1.2
Minneapolis/ St. Paul	1.3	5.2	2.8	2.9	1.9	2.1
Chicago	1.9	4.2	3.9	5.7	5.6	4.2
Albany	0.6	3.1	1.9	2.1	1.3	0.8

All figures are percentages. SMSA percentages for citations and controls are relative to *domestic* total.

It is not possible to tell for sure from these data when (if ever) that plateau tails off; the apparent tail-off in both panels of the figure is due at least in part to the 1989 observational cutoff.[19] For 1975 the higher citation rate for university patents is particularly pronounced in the early years; this pattern is not apparent in the 1980 cohort.

Before getting to our formal test of localization, an examination of Table II is useful to get a sense for the extent of geographic concentration in these data. It shows the fraction of patents coming from abroad and from a selection of major U. S. SMSAs for

19. The dropoff in both panels corresponds approximately to application year 1987 (1975 + 12 and 1980 + 7). Typically, a significant fraction of applications have not been granted within two years, so when we looked in 1989 this fraction of 1987 applications were not yet granted.

several of the datasets. Not surprisingly, a measurable fraction of university patents comes from Madison, WI; this is not true for corporate patents. A measurable (though smaller) fraction of the *citations* of university patents comes from Madison, and this fraction is larger than that for the controls. Indeed, the controls for the university citations look generally "more like" corporate patents than do the citations, suggesting that localization may be present. Other qualitative evidence of localization is apparent in the table, including the high percentage of NY SMSA citations that come from the NY SMSA.

The basic test of localization is presented in Table III. For each geographic area and each originating dataset, it presents the proportion of citations that geographically matched the originating patent. These proportions are shown both with and without self-citations. The matching proportions for the control samples are then shown, as well as a t-statistic testing the equality of the control proportions and the citation proportions (excluding self-citations).[20]

We focus first on the 1975 results on the left of the table. Starting with the country match, we find that citations *including self-citations* are domestic about 6 or 7 percent more often than the controls. Excluding self-citations eliminates this difference for the top corporate citations and cuts it roughly in half for the others. The remaining difference between the citations excluding self-cites and the controls is only marginally significant statistically.

Looking at the 1975 results for states, we find that citations of university patents come from the same state about 10 percent of the time; this rises to 15 percent for other corporate and 19 percent for top corporate. Excluding self-citations, however, makes a big difference. The university and top corporate proportions are cut to 6–7 percent, and the other corporate to just over 10. For the university and other corporate cohorts, the matching frequencies excluding self-citations are significantly greater than the matching control proportions.

20. Let p_c be the probability that a citation comes from the same geographic unit as the originating patent; let p_o be the corresponding probability for a randomly drawn patent in the same patent class (control). We test $H_o{:}p_c = p_o$ versus $H_a{:}p_c > p_o$ using the test statistic:

$$t = \frac{\hat{p}_c - \hat{p}_o}{\sqrt{[\hat{p}_c(1-\hat{p}_c) + \hat{p}_o(1-\hat{p}_o)]/n}},$$

where $\hat{p}_c$ and $\hat{p}_o$ are the sample proportion estimates of p_c and p_o. This statistic tests for the difference between two independently drawn binomial proportions; it is distributed as t.

TABLE III
GEOGRAPHIC MATCHING FRACTIONS

	1975 Originating cohort			1980 Originating cohort		
	University	Top corporate	Other corporate	University	Top corporate	Other corporate
Number of citations	1759	1235	1050	2046	1614	1210
		Matching by country				
Overall citation matching percentage	68.3	68.7	71.7	71.4	74.6	73.0
Citations excluding self-cites	66.5	62.9	69.5	69.3	68.9	70.4
Controls	62.8	63.1	66.3	58.5	60.0	59.6
t-statistic	2.28	−0.1	1.61	7.24	5.31	5.59
		Matching by state				
Overall citation matching percentage	10.4	18.9	15.4	16.3	27.3	18.4
Citations excluding self-cites	6.0	6.8	10.7	10.5	13.6	11.3
Controls	2.9	6.8	6.4	4.1	7.0	5.2
t-statistic	4.55	0.09	3.50	7.90	6.28	5.51
		Matching by SMSA				
Overall citation matching percentage	8.6	16.9	13.3	12.6	21.9	14.3
Citations excluding self-cites	4.3	4.5	8.7	6.9	8.8	7.0
Controls	1.0	1.3	1.2	1.1	3.6	2.3
t-statistic	6.43	4.80	8.24	9.57	6.28	5.52

Number of citations is less than in Table I because of missing geographic data for some patents. The t-statistic tests equality of the citation proportion excluding self-cites and the control proportion. See text for details.

At the SMSA level, 9 to 17 percent of total citations are localized. This again drops significantly when self-citations are excluded, but 4.3 percent of university citations, 4.5 percent of top corporate citations, and 8.7 percent of other corporate citations are localized excluding self-cites. This compares with control matching proportions of about 1 percent, and these differences are highly significant.

The results for citations of 1980 patents (right side of Table III) are even stronger and more significant. For every dataset, for every geographic level, the citations are quantitatively and statistically significantly more localized than the controls. The general increase in the proportion of U. S. patents taken by foreigners is reflected in a decline of 3 to 6 percent in the control percentages matching by country. The citation matching percentages actually rise, however, particularly for top corporate citations. It is impossible to tell from this comparison whether this represents a real change, or whether it is the result of the 1980 citations having shorter average citation lags. Since this gets to the issue of explaining which citations are localized, we postpone discussion until the next section.

Before moving on, the results on the extent of localization can be summarized as follows. For citations observed by 1989 of 1980 patents, there is a clear pattern of localization at the country, state, and SMSA levels. Citations are five to ten times as likely to come from the same SMSA as control patents; two to six times as likely excluding self-citations. They are three to four times as likely to come from the same state as the originating patent; roughly twice as likely excluding self-cites. Whereas about 60 percent of control patents are domestic, 70 to 75 percent of citations and 69 to 70 percent of citations excluding self-cites are domestic. Once self-cites are excluded, universities and firms have about the same domestic citation fraction; at the state and SMSA level there is weak evidence that university citations are less localized. For citations of 1975 patents, the same pattern, but weaker, emerges for citations of university and other corporate patents. For top corporate there is no evidence of localization at the state or country levels, though the SMSA fraction is significantly localized. Thus, we find significant evidence that citations are even more localized than one would expect based on the pre-existing concentration of technological activity, particularly in the early years after the originating patent.

III. Factors Affecting the Probability of Localization

The contrast between the 1975 and 1980 results suggests that localization of early citations is more likely than localization of later ones. This accords with intuition, since whatever advantages are created by geographic proximity for learning about the work of others should fade as the work is used and disseminated. Another hypothesis that is implicit in the previous discussion is that citations that represent research that is technologically similar to the originating research are more likely to be localized, because the individuals pursuing these related research lines may be localized. In addition, attributes of the originating invention or the institution that produced it may affect the probability that its spillovers are localized.

To explore these issues, we pooled the citations (excluding self-cites) to university and corporate patents for each cohort, and ran a probit estimation with geographic match/no match between the originating and citing patents as the dependent variable. As independent variables we included the log of the citation lag (set to zero for lags of zero), dummy variables for top corporate and other corporate originating patents, interactions of the lag and these dummies, and a dummy variable equal to unity if the citation has the same primary class as the originating patent. We also included a dummy variable that is unity if the *control* patent corresponding to this citation matches geographically with the originating patent, to control for the general increase over time in the fraction of U. S. patents granted to foreigners.

We also included two variables relating to the originating patent suggested by our work on basicness and appropriability of inventions [Trajtenberg, Henderson, and Jaffe, 1992]. The first, "generality" is one minus the Herfindahl index across patent classes of the citations received.[21] It attempts to capture the extent to which the technological "children" of an originating patent are diverse in terms of their own *technological* location. Thus, an originating patent with generality approaching one has citations that are very widely dispersed across patent classes; generality of zero corresponds to all citations in a single class. We argue elsewhere that generality is one aspect of the "basicness" of an invention. One might hypothesize that basic research results are less likely to be localized, because their spread is more likely to be through communication mechanisms (e.g., journals) that are not localized. The other variable characterizing the originating invention is the fraction of the originating patent's citations that were

21. Let C_{ik} be the number of citations received by patent i from subsequent patents whose primary patent class is k, and let C_i be the total number of citations received by patent i. The measure of generality is then

$$G_i = 1 - \sum_k \left(\frac{C_{ik}}{C_i}\right)^2.$$

TABLE IV
GEOGRAPHIC PROBIT RESULTS

	Country match		State match		SMSA match	
	1975	1980	1975	1980	1975	1980
Dummy for control	0.139	0.085	0.396	0.300	*	0.283
Sample match	(0.045)	(0.041)	(0.124)	(0.102)		(0.172)
Log of citation	−0.078	0.094	−0.264	0.198	−0.123	0.037
lag	(0.049)	(0.056)	(0.073)	(0.079)	(0.057)	(0.086)
Dummy for top	−0.114	−0.010	−0.383	0.013	−0.234	−0.208
corporate	(0.168)	(0.127)	(0.249)	(0.177)	(0.288)	(0.200)
Dummy for other	0.069	0.053	−0.214	−0.007	0.325	−0.042
corporate	(0.209)	(0.134)	(0.277)	(0.189)	(0.291)	(0.207)
Log-lag	0.046	−0.016	0.226	0.007	0.102	0.156
*top corp. dummy	(0.091)	(0.086)	(0.138)	(0.115)	(0.156)	(0.131)
Log-lag	0.008	−0.026	0.307	0.036	0.037	0.039
*other corp. dummy	(0.108)	(0.091)	(0.147)	(0.124)	(0.155)	(0.138)
Dummy for matching	−0.085	0.069	−0.013	0.034	−0.057	−0.016
patent class	(0.050)	(0.045)	(0.073)	(0.058)	(0.080)	(0.068)
Generality of origin	0.092	0.177	0.026	−0.140	0.013	−0.298
patent	(0.091)	(0.088)	(0.136)	(0.111)	(0.150)	(0.130)
Origin fraction	−0.813	0.162	0.815	0.883	1.174	0.828
Self-citations	(0.180)	(0.124)	(0.246)	(0.134)	(0.237)	(0.154)
# of observations	3581	4217	3573	4215	3566	3972
# of matches	2363	2925	256	490	197	298
Log likelihood	−2269	−2559	−894	−1459	−736	−1022

*The number of observations for which the control patent matched at the SMSA level was so small that this parameter could not be estimated.
Standard errors are in parentheses. All equations also included five technological field dummies.

self-cites. We take a high proportion of self-cites as evidence of relatively successful efforts by the original inventor to appropriate the invention. We expect that the nonself-citations to such a patent are more likely to be confined to suppliers, customers, or other firms that the inventing firm has a relationship with, and may therefore tend to be localized.

Finally, the extent of localization depends fundamentally on the mechanisms by which information flows, and these mechanisms may be different in different technical fields. For this reason, we also included dummy variables for broad technological fields.[22]

The results are presented in Table IV. Because of the presence

22. (1) Drugs and Medical Technology; (2) Chemicals and Chemical Processes Excluding Drugs; (3) Electronics, Optics, and Nuclear Technologies; (4) Mechanical Arts; and (5) All Other.

of the interaction terms between the lag and the corporate dummies, the coefficient on the lag itself corresponds to the fading of localization of citations of university patents. There is evidence in the 1975 results of such fading. This effect is statistically significant at the state and SMSA levels; its quantitative significance is discussed further below. For the citations of corporate patents, the interaction terms measure the difference between their fading rates and those of university citations. These terms are generally not statistically significant. In only one case (other corporate, 1975) could we reject the hypothesis of equality of fading rates at traditional confidence levels. There is, however, weak evidence that the corporate citations do not fade as rapidly as those of university patents, at least at the state and SMSA levels. The coefficients on the corporate dummies themselves capture differences in the predicted probability of localization for citations with lags of zero or one year. These are all insignificant, and there is no clear pattern.

The matching-patent-class and generality variables do not work well. The effects are generally insignificant, and show no consistent pattern. The effect of the self-citation fraction, however, is strong and puzzling. At the state and local level, there is a very significant effect in the predicted direction: citations of patents with a high self-citation fraction are more likely to be localized. This is *not* just saying that self-citations are localized, since they are excluded; it is the other citations that are more localized. At the country level, at least in 1975, this effect is reversed and is significant. Taking all results together, it suggests that for patents with a lot of self-citations, the nonself-citations are more likely to be foreign, but those that are domestic are more likely to be in the same state and SMSA as the originating patent.

The 1980 results are disappointing. The coefficient on the time lag term switches sign, though it is generally insignificant. One possibility is that these citations span too short a time period to capture the lag effect well. To test this possibility, we reran the estimation in Table IV on the 1975 citations, excluding all that were granted after 1984. This analysis tells us what we would have believed about citations of 1975 patents if we had looked for them only as long as we have looked for the citations of the 1980 patents. The results (not reported here) looked "more like" the 1980 results than the original 1975 results did. In particular, the coefficient on the lag term was insignificant, and was positive at the SMSA level.

TABLE V
PREDICTED LOCALIZATION PERCENTAGES OVER TIME (BASED ON 1975 PROBIT RESULTS FOR CITATIONS OF UNIVERSITY PATENTS)

	Predicted percentage for:		
Lag	Same country	Same state	Same SMSA
0 or 1 year	67.1	9.7	4.8
5 years	65.5	6.5	4.0
10 years	64.6	5.3	3.7
25 years	63.5	4.0	3.3

Thus, it may be that the "perverse" results for the 1980 sample would go away if we had later citations to include.

A probit coefficient does not have an economically meaningful magnitude, because of the need to standardize the variance of the underlying error distribution. However, we can calculate what the coefficients imply about changes in the predicted probabilities. This is done in Table V, using the 1975 lag coefficient.[23] Table V was constructed by calculating the predicted localization probability using the results of Table IV, evaluating the citation lag at different values, and evaluating the other independent variables at the mean of the data. It shows that the estimates correspond to a reduction in the localization fraction after, for example, ten years, from 67.1 percent to 64.6 percent at the country level, 9.7 percent to 5.3 percent at the state level, and 4.8 percent to 3.7 percent at the SMSA level.

IV. DISCUSSION AND CONCLUSION

Despite the invisibility of knowledge spillovers, they do leave a paper trail in the form of citations. We find evidence that these trails, at least, are geographically localized. The results, particularly for the 1980 cohort, suggest that these effects are quite large and quite significant statistically. Because of our interest in true externalities, we have focused on citations excluding self-cites. For some purposes, however, this is probably overly conservative. From the point of view of the Regional Development Administra-

23. As discussed above, this is the point estimate of the lag coefficient for citations of university patents. The point estimates are different for the corporate originating patents, but we have not performed separate calculations for each dataset.

tor, it may not matter whether the subsequent development that flows from an invention is performed by the inventing firm, as long as it is performed in her state or city. Our results are also conservative because we attribute none of the localization present in the control samples to spillovers, despite the likelihood that spillovers are, indeed, one of the major reasons for the pre-existing concentration of research activity.

We also find evidence that geographic localization fades over time. The 1980 citations, which have shorter average citation lags, are systematically more localized than the 1975 citations. By using a probit analysis, we produced estimates of the rate of fading. These estimates seem to suggest a rate of fading that is both smaller than one would expect, and smaller than would be necessary to explain the difference between the 1975 and 1980 overall matching fractions. One possibility is that the difficulty of measuring the rate of fading is due to the "contamination" of citations by the patent examiner. As noted in footnote 18, it is particularly likely that citations with very short lags were added by the examiner. If we believe that such citations are less likely to represent spillovers and less likely to be localized, then this would tend to bias toward zero our measure of the effect of time on localization.[24]

We find less evidence of the effect of technological area on the localization process. Citations in the same class are no more likely to be localized. Overall, there is not really any evidence in these data that the probability of coming from a given geographic location conditional on patent class is different from the unconditional probability. This may be due to the arbitrary use of the "primary" patent class, to the exclusion of the "cross-referenced" classes. There is no legal difference in significance between the primary and cross-referenced classes, and in many cases the examiners do not place any significance on which class is designated primary. In future work we hope to explore whether using the full range of information contained in the cross-referenced classes provides a better technological characterization of the patents.

In this context it is worth noting that part of what is going on is probably that knowledge spillovers are not confined to closely related regions of technology space. Approximately 40 percent of

24. Attempts to test for this possibility by including a dummy variable for age 0 or 1, as well as other explorations of possible nonlinearity in the match-lag relationship, were inconclusive.

citations do not come from the same primary patent class; even at the level of the five broad "technological fields" listed in footnote 21, 12 to 25 of percent of citations are across fields. This is consistent with Jaffe [1986], which found that a significant fraction of the total "flow" of spillovers affecting firms' own research productivity comes from firms outside of the receiving firm's immediate technological neighborhood.

We find surprisingly little evidence of differences in localization between the citations of university and corporate patents. The largest difference is that corporate patents are more often self-cited, and self-cites are more often localized. The probit results do not allow rejection of the hypothesis that the initial localization rates for nonself-citations are indistinguishable for the different groups. They do provide some weak evidence that this initial localization is more likely to fade for the university patents, at least at the state and local levels.

In order to provide a true foundation for public policy and economic theorizing, we would ultimately like to be able to say more about the mechanisms of knowledge transfer, and about something resembling social rates of return at different levels of geographic aggregation. The limitations of patent and citation data make it difficult to go much farther with such questions within this research approach. Ex post, the vast majority of patents are seen to generate negligible private (and probably social) returns. In future work we plan to identify a small number of patents that are extremely highly cited. It is likely that such patents are both technologically and economically important [Trajtenberg, 1990]. Case studies of such patents and their citations could prove highly informative about both the mechanisms of knowledge transfer, and the extent to which citations do indeed correspond to externalities in an economic sense.

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REFERENCES

Bernstein, J., and M. Nadiri, "Interindustry R&D Spillovers, Rates of Return, and Production in High-Tech Industries," *American Economic Review Papers and Proceedings,* LXXVIII (May 1988), 429–34.

Bernstein, J., and M. Nadiri, "Research and Development and Intra-industry Spillovers: An Empirical Application of Dynamic Duality," *Review of Economic Studies,* LVI (April 1989), 249–67.

Dorfman, N., "Route 128: The Development of a Regional High-Technology

Economy," D. Lampe, ed., *The Massachusetts Miracle: High Technology and Economic Revitalization* (Cambridge, MA: MIT Press, 1988).
Feller, Irwin, "R&D Theories and State Advanced Technology Programs," Paper prepared for the American Association for the Advancement of Science Annual Meeting, January 1989.
Glaeser, Edward, H. D. Kallal, J. A. Scheinkman, and A. Shleifer, "Growth in Cities," NBER Working Paper No. 3787, July 1991.
Griliches, Z., "The Search for R&D Spillovers," NBER Working Paper No. 3768, July 1991.
Grossman, G., and E. Helpman, *Innovation and Growth in the Global Economy* (Cambridge, MA: M.I.T. Press, 1991).
Jaffe, Adam, "Technological Opportunity and Spillovers of R&D: Evidence from Firms' Patents, Profits and Market Value," *American Economic Review*, LXXVI (1986), 984–1001.
——, "Real Effects of Academic Research," *American Economic Review*, LXXIX (1989), 957–70.
Krugman, P., *Geography and Trade* (Cambridge: M.I.T. Press, 1991).
Mansfield, E., "Sources and Characteristics of Academic Research Underlying Industrial Innovations," mimeo, University of Pennsylvania, 1991.
Marshall, A., *Principles of Economics* (London: Macmillan, 1920).
Minnesota Department of Trade and Economic Development, Office of Science and Technology, *State Technology Programs in the United States*, 1988.
Romer, P., "Increasing Returns and Long-Run Growth," *Journal of Political Economy*, XCIV (1986), 1002–37.
——, "Endogenous Technological Change," *Journal of Political Economy*, XCVIII (1990), S71–S102.
Smilor, R., G. Kozmetsky, and D. Gibson, *Creating the Technopolis: Linking Technology, Commercialization and Economic Development* (Cambridge, MA: Ballinger Publishing Co., 1988).
Trajtenberg, M., "A Penny for Your Quotes: Patent Citations and the Value of Innovations," *Rand Journal of Economics*, XXI (1990), 172–87.
Trajtenberg, M., R. Henderson, and A. Jaffe: "Ivory Tower Versus Corporate Lab: An Empirical Study of Basicness and Appropriability," NBER Working Paper No. 4146, 1992.

CORRUPTION*

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This paper presents two propositions about corruption. First, the structure of government institutions and of the political process are very important determinants of the level of corruption. In particular, weak governments that do not control their agencies experience very high corruption levels. Second, the illegality of corruption and the need for secrecy make it much more distortionary and costly than its sister activity, taxation. These results may explain why, in some less developed countries, corruption is so high and so costly to development.

I. INTRODUCTION

We define government corruption as the sale by government officials of government property for personal gain. For example, government officials often collect bribes for providing permits and licenses, for giving passage through customs, or for prohibiting the entry of competitors. In these cases they charge personally for goods that the state officially owns. In most cases the goods that the government officials sell are not demanded for their own sake, but rather enable private agents to pursue economic activity they could not pursue otherwise. Licenses, permits, passports, and visas are needed to comply with laws and regulations that restrict private economic activity. Insofar as government officials have discretion over the provision of these goods, they can collect bribes from private agents.

Corruption is both pervasive and significant around the world. In some developing countries, such as Zaire and Kenya, it probably amounts to a large fraction of the Gross National Product. Corruption is also common in the developed countries: defense officials sometimes sell contracts for personal gain, and local zoning officials are bribed to rezone. Still, economic studies of corruption are rather limited. Following Becker and Stigler [1974], most studies (e.g., Banfield [1975], Rose-Ackerman [1975, 1978], and Klitgaard [1988, 1991]), focus on the principal-agent model of corruption. This model focuses on the relationship between the principal, i.e., the top level of government, and the agent, i.e., an official, who takes the bribes from the private individuals interested in some government-produced good. These studies examine ways of motivating the agent to be honest, ranging from efficiency

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wages [Becker and Stigler, 1974] to indoctrination [Klitgaard, 1991]. In this paper we take the principal-agent problem as given—the corrupt official has some effective property rights over the government good he is allocating—and focus on *consequences* of corruption for resource allocation.

In particular, we address two issues. First, we discuss the implications of how the corruption network is organized. In some economies, such as Korea today and Russia under Communists, while corruption is pervasive, the person paying the bribe is assured that he gets the government good that he is paying for, and does not need to pay further bribes in the future. In other economies many government goods can be obtained without bribes altogether. For example, a citizen can get a passport in the United States without paying a bribe. In yet other economies, such as many African countries and post-Communist Russia, numerous bureaucrats need to be bribed to get a government permit, and bribing one does not guarantee that some other bureaucrat or even the first one does not demand another bribe. We examine the implications of these three regimes for the level of corruption and for the effects of corruption on economic activity.

Second, we ask why even well-organized corruption appears to be more distortionary than taxation. Several authors have pointed out that some corruption might be desirable [Leff, 1964]. First, it works like a piece rate for government employees (a bureaucrat might be more helpful when paid directly). Second, it enables entrepreneurs to overcome cumbersome regulations. Yet most studies conclude that corruption slows down development [Gould and Amaro-Reyes, 1983; United Nations, 1989; and Klitgaard, 1991]. We ask why bribery might be much more costly than its sister activity, taxation, and argue that the imperative of *secrecy* makes bribes more distortionary than taxes.

The next section sets out our basic model of corruption, and briefly addresses the question of why corruption spreads. Section III looks at the market structure of the supply of government goods as a determinant of the level and consequences of corruption. Section IV examines the costs of corruption focusing on secrecy. Section V concludes.

II. Basic Model

To fix ideas, we consider the simplest model of one government-produced good, such as a passport, or a right to use a government road, or an import license. We assume that this good is homogeneous, and that there is a demand curve for this good, $D(p)$, from the private agents. We assume that this good is sold for the government by an official, *who has the opportunity to restrict the quantity of the good that is sold.* Specifically, he can deny a private agent the passport, access to a road, or an import license. In practice, this denial might mean a long delay or an imposition of many requirements. But it is easier to assume for now that the official can simply refuse to provide the good. An important reason why many of these permits and regulations exist is probably to give officials the power to deny them and to collect bribes in return for providing the permits [De Soto, 1989].

We also assume that the official can in fact restrict supply without any risk of detection or punishment from above. Corrupt officials go unpunished because their bosses often share in the proceeds and because public pressure to stop corruption in most countries is weak. We shall also discuss the case in which corruption is penalized. But for now, the government official is a monopolist selling the good. His objective is to maximize the value of the bribes he collects from selling this government good.

Let the official government price of this good be p. We assume that the cost of producing this good is completely immaterial to the official since the government is paying this cost. This assumption is a bit restrictive. While it covers the sale of an import license, a passport, or a passage on a government road, a policeman who sells his services that he is supposed to provide for free does exert personal effort and so does care about its cost. For simplicity, we focus on government goods that cost the official nothing personally to provide, so that he has no interest in how much it costs the government to produce these goods.

What then is the marginal cost to the official of providing this good? We distinguish two cases. First, in the case *without theft,* the official actually turns over the official price of the good to the government. In this case, the marginal cost of providing the good to the official is this government price p. For example, when an official sells a license for a government price plus a bribe, he keeps the bribe but the amount p stays with the government; hence p is his marginal cost. In contrast, in the case *with theft,* the official does not turn over anything to the government at all, and simply hides the sale. In this case, the price that the buyer pays is only equal to the bribe, and might be even lower than the official price. For example, customs officials often let goods through the border for less than the official duty, but then give nothing at all to the government. In this case, the marginal cost to the official is zero. While conceptually the two cases are similar—they differ only in the level of the marginal cost to the official—in the first case corruption always raises the total price of the good, whereas in the second case it might reduce it. Corruption with theft is obviously more attractive to the buyers.

If the official cannot price discriminate between buyers, then as a monopolist, he will simply set the marginal revenue equal to the marginal cost. In the case without theft, the total price with the bribe always exceeds the government price. It pays the official to create a shortage at the official price, and then to collect bribes as a way to clear the market for the government-supplied good [Shleifer and Vishny, 1992]. In the case with theft, the total price might be below the government price. Figures Ia and Ib present the solutions to this problem for the cases without and with theft, respectively.

This analysis suggests a similarity between bribes and commodity taxes. In the case without theft, the bribe is exactly equal to the revenue-maximizing commodity tax when marginal cost is equal to the state price p. Of course, taxes need not be set to maximize revenue. More importantly, taxes are typically kept by the government rather than the bureaucrats. In monarchal regimes, the

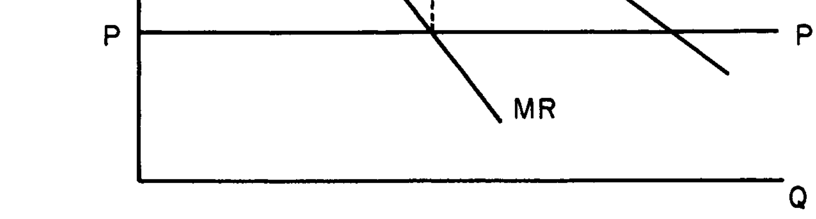

FIGURE Ia
Corruption without Theft

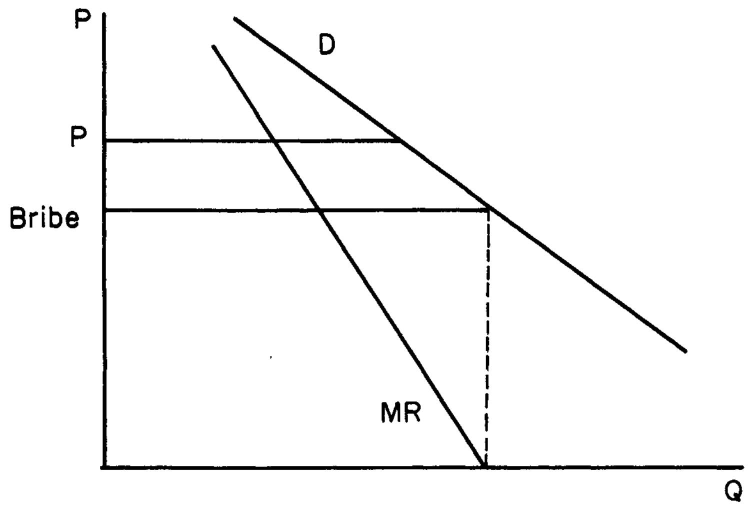

FIGURE Ib
Corruption with Theft

distinction between taxes and bribes is blurred by the fact that the treasury is indistinguishable from the sovereign's pocket. Yet for most governments, the distinction is material and shows how corruption substitutes for taxation.

Penalizing the official for corruption changes the level of the bribe he demands, but does not change the essence of the problem. If the probability of detection and the penalty are independent of the bribe and of the number of people who pay it, the official will charge the same bribe provided that the penalties are not so high that corruption is no longer profitable. If the expected penalty increases with the level of the bribe, he might reduce the bribe and raise output. On the other hand, if the expected penalty rises in the number of people he charges a bribe (for example, because of the higher probability of a complaint), then he will reduce the supply and raise the bribe. The official trades off the benefits given in Figures Ia and Ib against the expected penalties. For our purposes, we do not need to focus on this aspect of the problem (see Becker and Stigler [1974], Rose-Ackerman [1978], and Klitgaard [1988]).

This simple analysis suggests that corruption spreads because of competition both between the officials and between the consumers. If jobs are distributed among officials through an auction mechanism, whereby those who pay the most for a job get it, then the prospective officials who do not collect bribes simply cannot afford jobs. Conversely, those who will collect more (perhaps through more effective price discrimination), will offer the higher officials more for the jobs, and so will be able to get them. Competition between officials will assure that maximal bribes are collected.

Even more important for the spread of corruption is competition between the buyers in the case with theft. If buyer A can buy the government service more cheaply than buyer B can, then he can outcompete buyer B in the product market. So if buyer A bribes an official to reduce his costs, his competitors must do so also. If all real estate owners in a city can bribe their way out of paying taxes, then those who pay them will not survive. If some trucks carry goods across a border after paying a small bribe instead of the official customs duty, the importers who pay the duty will not survive. Competition between buyers of government services assures the spread of cost-reducing corruption. Interestingly, such competition does not help the spread of corruption *without* theft.

Corruption with theft spreads because observance of law does not survive in a competitive environment. In addition, the buyer in this case has no incentive to inform on the official, and hence the likelihood that corruption is detected is much smaller. This creates a further incentive for corruption with theft to rise. Because corruption with theft aligns the interests of the buyers and sellers, it will be more persistent than corruption without theft, which pits buyers against sellers. This result suggests that the first step to reduce corruption should be to create an accounting system that prevents theft from the government. In the collection of taxes and customs duties, such accounting systems might well reduce corruption because without theft bribes raise the buyer's cost and hence give him the incentive to expose the corrupt official.

III. The Industrial Organization of Corruption

The model above makes two strong assumptions. First, a buyer needs only one government good to conduct his business. Second, the official is a monopolist in the supply of this good. Yet some critical issues in corruption arise when these assumptions do not hold. In many cases, a private agent needs several *complementary* government goods to conduct business. For example, an importer might need several government licenses and permits, to be obtained from several agencies, to bring in, unload, transport, and sell an imported good. A builder might need several permits from different departments, such as fire, water, and police. With multiple goods, the market structure in their provision becomes important. The different agencies that supply the complementary goods might collude, sell the different goods independently, or even compete in the provision of some goods. The focus on market structure in the provision of complementary government goods sheds light on the consequences of corruption.

The model of the previous section is most appropriate for understanding corruption in monarchies, such as the Bourbons in France or Marcos' Philippines, in the old-time Communist regimes, and in regions dominated by a single mafia. In such places, it is always clear who needs to be bribed and by how much. The bribe is then divided between all the relevant government bureaucrats, who agree not to demand further bribes from the buyer of the package of government goods, such as permits. In Russia, for example, bribes were channeled through local Communist party offices. Any deviation from the agreed-upon pattern of corruption would be penalized by the party bureaucracy, so few deviations occurred. Once a bribe was paid, the buyer got full property rights over the set of government goods that he bought. Carino [1986] and Klitgaard [1988] describe similar monopolistic corruption structures in the Philippines.

There are two extreme alternatives to this monopoly corruption scheme. The first alternative is corruption in some African countries, in India, and in post-Communist Russia. Here the sellers of the complementary government goods, such as permits and licenses, act independently. Different ministries, agencies, and levels of local government all set their own bribes independently in an attempt to maximize their own revenue, rather than the combined revenue of all the bribe collectors. In Russia in 1991, for example, getting a business started often required bribing the local legislature, the central ministry, the local executive branch, the fire authorities, the water authorities etc. In some African countries, many quasi-independent government agencies have the power to stop a project, and use it to set bribes without collusion with other agencies [Klitgaard, 1990]. The army and the police also often demand a cut for protection—another needed government input. Unlike the single monopoly model, here complementary government goods are sold by independent monopolists.

Formally, consider first a joint monopolist agency that sets the cum bribe prices p_1 and p_2 of two government goods. Let x_1 and x_2 be the quantities of these goods sold. Let the official prices, equal to the monopolist's marginal costs, be denoted by MC_1 and MC_2. The per unit bribes then are $p_1 - MC_1$ and $p_2 - MC_2$. The joint monopolist agency sets p_1 at which

$$MR_1 + MR_2 \frac{dx_2}{dx_1} = MC_1, \tag{1}$$

where MR_1 and MR_2 denote marginal revenues from the sale of goods 1 and 2, respectively. When the two goods are complements, as government permits for the same project are, then $dx_2/dx_1 > 0$, and so at the optimum, $MR_1 < MC_1$. The monopolist agency keeps the bribe on good 1 down to expand the demand for the complementary good 2 and thus to raise its profits from bribes on good 2. For the same reason, this agency keeps down the price of good 2.

Suppose alternatively that permits 1 and 2 are allocated by independent agencies. Each agency then takes the other's output as given, and in particular, in equation (1), dx_2/dx_1 is set to zero. At the independent agency's optimum, $MR_1 = MC_1$. Hence the per unit bribe is higher, and the output lower, than at the joint monopolist optimum. Because the independent agency ignores the effect of its raising its bribe on demand for the complementary permits and hence the bribes to the other agency, it sets a higher bribe, which results in a lower output and a lower aggregate level of bribes. By acting independently, the two agencies actually hurt each other, as well as the private buyers of the permits.

This problem is made much worse in many countries by free entry into the collection of bribes. New government organizations and officials often have the opportunity to create laws and regulations that enable them to become providers of additional required permits and licenses and charge for them accordingly. Having paid three bribes, the buyer of these inputs learns that he must buy yet another one if he wants his project to proceed. In some cases, the officials who have collected the bribe previously come back to demand more (see Klitgaard [1990] for striking examples). In these cases, the property rights to his project are not really transferred to the buyer when he pays the bribe. The point is that even the list of the complementary inputs is not fixed, and tends to expand when profitable corruption opportunities stimulate entry. When entry is completely free, the total bribe rises to infinity and the sales of the package of government goods, as well as bribe revenues, fall to zero.

In the third scenario, *each one* of the several complementary government goods can be supplied by at least two government agencies. For simplicity, begin with the case of one such good, such as a United States passport or a driver's license. A citizen can obtain a U. S. passport without paying a bribe. The likely reason for this is that if an official asks him for a bribe, he will go to another window or another city. Because collusion between several agents is difficult, bribe competition between the providers will drive the level of bribes down to zero. This example can be extended to the case of multiple complementary goods. If a builder needs several permits to erect a building, but any one of them can be obtained from one of several noncolluding government agents, Bertrand competition in bribes will force the equilibrium bribe on each permit down to zero. Unlike the first model, where a unified monopoly provides all the goods, and the second model, in which monopoly suppliers of different goods act independently, here the market for each government-supplied good is competitive.

As in other industrial organization contexts, even having two competitors is not necessary if the market is subject to potential competition or entry [Demsetz, 1968]. Consider, for example, a single government employee in a small U. S. city, who controls building permits, dog permits, permits to dispose of old appliances, etc. If this employee attempts to charge a bribe, or to price his services above marginal cost, another individual would offer the public the same service at a lower price, and the corrupt official will be recalled or fail to get reelected. The threat of such competition would then keep corruption down to zero, assuming that the official price covers the marginal cost of providing the permits.

The level of bribes is the lowest in the third case, intermediate in the first, and the highest in the second. But the total amount of revenues collected is higher in the first case than in the second, since the independent monopolist suppliers drive the quantity sold so far down that the total revenues from corruption fall. This result is obvious: in the first case the suppliers of the complementary inputs collude to maximize the total value of bribes, but in the second they do not.

This problem is formally identical to a standard problem in industrial organization. Suppose that a carmaker needs two complementary inputs, glass and steel. If both are provided by one monopolist, he will realize that raising the price of glass reduces the demand for his own steel, and hence his profits on the steel sales, and similarly with raising the price of steel. Accordingly, he will price steel and glass taking account of the demand complementarities. In contrast, if glass and steel are sold by two independent monopolists, each will ignore the effect of his raising his price on the demand for the product of the other. As a result, each would charge a higher price than a joint monopolist would, and both the quantity of steel and glass sold, and the combined profits from these sales would be lower. In the last scenario, if each of these independent monopolists can sell both steel and glass, and they compete on price, they will drive the price of both steel and glass down to the marginal cost. The profits will be the lowest, and output the highest, of the three cases. Competition is the best; joint monopoly is the second best; and independent monopoly is the worst for efficiency. Moreover, the more inputs car production requires, the lower is output with independent monopolists.

Another helpful analogy is to tollbooths on a road. The joint monopoly solution corresponds to the case of one toll that gives the payer the right to use the entire road. The independent monopolists solution means that different towns through which the road passes independently erect their own tollbooths and charge their own tolls. The volume of traffic and aggregate toll collections fall. In fact, they fall to zero when *any* party can erect its own tollbooth on this road. The competitive case corresponds to multiple booths competing with each other for the right to collect the toll, or alternatively to the case of multiple roads. In this case, the volume of traffic is obviously the highest, and toll collections are the lowest.

This, in fact, is a very close analogy. In India, taking a road between two towns indeed requires paying a bribe in every village through which the road passes. Taking goods inland in Zaire is more expensive because of corruption than bringing them from Europe by ship to a port. In 1400 there were 60 independently run tolls along the Rhine. Along the Seine there were so many tolls that to ship a good twenty miles cost as much as its price. In contrast, rivers in England were free of such tolls, which in part explains the ability of England to develop specialized, commercial agriculture feeding London, the world's center of commerce [Heilbroner, 1962]. These examples suggest how costly free entry into bribe collection might be to development.

This industrial organization perspective on corruption sheds light on the consequences of corruption in different countries and places. It also raises the far deeper question: what determines the industrial organization of the different corruption markets? How did Brezhnev and Marcos manage to enforce joint profit maximization? Why has this system fallen apart in Russia, and never existed in Africa? How has the U. S. government managed to eradicate corruption in the provision of at least some, though by no means all, government goods?

Enforcement of joint profit maximization in bribe collection is closely related to the problem of enforcing collusion in oligopoly. Stigler [1964] shows that collusion is more likely to be enforced when price-cutting can be easily detected, and punishment for price-cutting can be severe. In the corruption context the parallel argument is that collusive bribe maximization can be enforced more easily when bribe *increases* can be more easily detected and more severely punished.

Bribe increases can be easily detected in several circumstances. First, when the government has an effective policing machine to monitor the actions of the bureaucrats, such as the KGB in the Soviet Union or Mayor Daley's Democratic Party machine in Chicago, it is hard to charge excessive bribes without being found out. Second, when the ruling elite is small, as in the Philippines or in Communist Russia, deviations from normal bribes will be easy to see. Third, when the society is homogeneous and closely knit, as in East Asia, deviations from normal bribes are likely to become known to friends and family, and such knowledge is likely to spread. Police states, small oligarchies, and homogeneous societies are thus likely to come closer to joint bribe maximization than more open, less tightly governed and more heterogeneous societies.

The ability of the cartel to punish those who charge excessive bribes is also essential to enforcing collusion. The ability of the leadership to exclude deviators from the rents associated with being an insider is essential. When large rents come from being a communist in Russia, a democratic politician in Chicago, a part of the ruling clique, or a member of the military elite, and when the sovereign can take these rents away from the deviators, deviations are unlikely. On the other hand, if the rents are small, and, more importantly, the sovereign is in no position to take them away, joint bribe maximization cannot be sustained. For example, in feudal Europe, in post-Communist Russia, and in many African countries, the central government is so weak that it cannot fire or penalize officials in the provinces, or even bureaucrats sitting in the capital, for running their own corruption rackets. In this situation the "independent monopolists" model, with its devastating economic consequences, describes reality best.

Huntington [1968] observes that political modernization, defined as a transition from an autocratic to a more democratic government, is usually accompanied by increases in corruption. He attributes this problem to underdeveloped institutions under the newly formed governments. If underdeveloped institutions mean a weak state machine, then Huntington's story fits well with our model. New governments lose monopoly over bribe collection, and as a result, multiple agencies take bribes where only one did before, leading to a much less efficient allocation. In the Philippines under Marcos, all corruption flowed to the top; since his demise, the number of independent bribe takers has increased, and so the efficiency of resource allocation has probably declined. Russia under Communists had a monolithic bribe collection system. With Communists gone, central government officials, local officials, ministry officials, and many others are taking bribes, leading to much higher bribes in equilibrium though perhaps lower corruption revenues, just as the model predicts. Similar stories are told about Africa after independence, when the colonial corruption machines disintegrated [Ekpo, 1979]. The evidence is strikingly consistent in showing the superiority of monopolistic bribe taking over that by independent monopolists.

The two cases we examined share basically authoritarian governments with little responsiveness to public pressure against corruption. As a result, both produce high levels of corruption, although they differ in how inefficient this corruption is. Countries with more political competition have stronger public pressure against corruption—through laws, democratic elections, and even the independent press—and so are more likely to use government organizations that contain rather than maximize corruption proceeds. It is implausible to think, for example, that the U. S. president maximizes corruption proceeds, since such a president is likely to be exposed and thrown out of office. Even in Japan and Korea, where corruption is very common, the level of bribes tends to be significantly lower than in Russia or the Philippines. The likely reason for this is political competition within the ruling parties as well as from the opposition parties in these countries. Because low bribes keep potential competitors out, political competition keeps corruption down (see Demsetz [1968]).

Our industrial organization perspective suggests that the best arrangement to reduce corruption *without theft* is to produce competition between bureaucrats in the provision of government goods, which will drive bribes down to zero. The passport office, and many other agencies of the U. S. government, have actually introduced such arrangements. The Pentagon has not, and it is probably more corrupt. The general idea behind federalism is precisely such competition in the provision of public goods, although it is usually stated in terms of taxes rather than bribes. Of course, in the case of corruption *with theft,* competitive pressure might increase theft from the government at the same time as it reduces bribes. The appropriate policy, then, is to create competition in the provision of government goods while intensively monitoring theft.

IV. Corruption and Secrecy

Although some political scientists have argued that the optimal level of corruption is positive [Leff, 1964; Huntington, 1968], most studies suggest that existing corruption levels are detrimental to development [Gould and Amaro-Reyes, 1983; United Nations, 1989; Klitgaard, 1991]. Africa is reputed to be a very corrupt continent; it is also the poorest one. Central and South America are also known for the extreme corruption and poverty. In contrast, developed countries appear to be less corrupt.

Mauro [1993] presents the first systematic empirical analysis of corruption by focusing on the relationship between investment and corruption. Mauro uses an index of corruption from *Business International* [1984], a publication of *Economist Intelligence Unit,* which supplies subjective assessments of 56 risk factors for 68 countries to private investors. The corruption variable is defined as "the degree to which business transactions involve corruption and questionable payments," and is used for 1980. The average ratio of total and private investment to GDP for the period between 1970 and 1985 is drawn from Barro [1991], as is real GDP per capita for 1980. Mauro finds that, holding 1980 real GDP constant, countries with higher corruption have a lower ratio of both total and private investment to GDP. The estimates are statistically significant. These results are consistent with the view that corruption is bad for development.

The independent monopolists model, which shows that under free entry of bribe takers supplying complementary inputs the total bribe rises to infinity and productive output falls to zero, may help explain why the most corrupt countries are so poor. Yet even more modest corruption seems to have detrimental effects. In this section we discuss these detrimental effects of corruption.

In the case of an economywide bribe-collecting monopolist, such as Marcos, corruption is similar to revenue-maximizing taxation. Like the sovereign who optimally taxes different goods and activities, the monopolist will set bribes to maximize revenue. In this world it is difficult to distinguish between bribes and taxes. Taxes are the markup on the price that goes into the treasury, and bribes are the markup that goes into the pocket of the monopolist. When the treasury and the pocket are one and the same, as in the case of kings and Marcos, taxes and bribes are exactly the same. With multiple monopolists, bribes are also similar to taxes, except that tax rates on different activities are set by independent agencies. In setting tax rates in this way, the agencies maximize their own tax revenues rather than the aggregate tax revenue. Because they ignore the cross elasticities of demand, the aggregate tax revenues are lower in this case. Finally, the case of competing monopolists corresponds to the federalist ideal of competing jurisdictions. In this case as well, bribes are similar to taxes.[1]

Despite these similarities, bribes differ from taxes in one crucial way, namely, unlike taxation, corruption is usually illegal and must be kept secret. Efforts to avoid detection and punishment cause corruption to be more distortionary than taxation. On some goods, taking bribes without being detected is much easier than on others. Government officials will then use their powers to induce substitution into the goods on which bribes can be more easily collected without detection. For example, officials might ban some imports to induce substitution into others. Or they might prohibit entry of some firms to raise bribe revenue from existing monopolies. Historically, sovereigns used such mercantilist policies to increase tax collections because monopoly profits are easier to tax than income [Ekelund and Tollison, 1981]. But such policies can also be used to increase bribes. Using our roadblock analogy, bureaucrats shut down some roads to increase the tolls on the passage through others, especially if the tolls on the shut-down roads are more difficult to collect.

A very simple numerical example may clarify this point. Suppose that a country can import either green or red cars, and that the border price of either car is 5. Suppose that consumers demand only ten cars total and that the valuation of a red car is 15 for each consumer but of a green car it is only 10. In a free market the country will import only red cars at the price of 5, and end up with a consumer surplus of $10 \times (15 - 5) = 100$. If the ministry

1. Importantly, if corruption with theft *replaces* taxes, then the corrupt state might have to replace the lost revenue through very distortionary taxation.

could tax car imports, it would charge an import duty of 10 per car, which would result in the importation of ten red cars, no consumer surplus, and the government revenue of 100. In this case, taxes lead to no efficiency losses but a redistribution from consumers to the government. Suppose alternatively that the trade ministry bureaucrats want to raise revenue through bribes rather than taxes. However, they cannot undetectably collect bribes at the border for importing *red* cars (which are too bright and noticeable), but can collect bribes for importing green cars. The ministry then bans red car imports altogether, and demands a bribe of $10 - 5 = 5$ on each imported green car. In equilibrium, no red cars are imported, the consumer surplus falls to zero, and bureaucrats collect $10 \times 5 = 50$ on the import of green cars. Social surplus falls from 100 in the case of taxation to 50 in the case of corruption.

The surplus is even lower if resources are spent by the bureaucrats on securing their positions, and by them and importers on avoiding detection and punishment. These rent-seeking activities consume resources and dissipate gains from bribes [Tullock, 1967; Krueger, 1974]. In the extreme case, the cost of such rent-seeking activities adds up to the whole remaining bribe surplus of 50. In this case, corruption eliminates the social surplus from imports completely.

A real-world example of a bottle-making factory in Mozambique illustrates these distortions from corruption. In 1991 that factory had modern Western equipment for making bottles, but used a traditional process for putting paper labels on these bottles. Three old machines were used: one cut the labels from paper; one then glued the white label on the bottle; and finally one printed a red picture on the label. The bottles were moved manually between these machines. In roughly 30 percent of the cases, the picture was not centered on the label. When this happened, the bottles were handed over to approximately twelve women who sat on the floor near the machines and scraped off the labels with knives, so that the bottles could be put through this process again.

Apparently, the process of labeling bottles could be mechanized with a fairly simple machine that cost about $10,000 and could be readily bought with aid money from any of a number of western or even Third World suppliers. The manager of the factory, however, did not want to buy such a machine, but instead wanted to have a $100,000 machine, that not only mechanized the existing process, but also printed labels in sixteen colors and different shapes, and put them on different types of bottles. Only one producer in the world made that machine, and the Mozambiquan government applied to the producer's home country for an aid package to buy it. Since that aid was not immediately forthcoming, the factory kept using the traditional technology.

The demand for equipment much fancier than the factory appeared to need seems irrational until one realizes that buying a fancier machine offered the manager (and the ministry officials) much better opportunities for corruption. If the factory bought a generic machine, the manager would probably have to use international donors' guidelines and consider several offers. There would be very little in this deal for him personally. On the other hand, if he got a unique machine, he would not have to solicit alternative bids. The supplier in turn would be happy to overinvoice for the machine, and kick back some of the profits to the manager (and his ministerial counterpart). The corruption opportunities on buying a unique and expensive machine are much better than such opportunities on buying cheaper generic products.

The social cost of corruption in this example may be large. If the social value of the \$100,000 machine is only \$20,000, and the bribe that the manager can collect from overinvoicing is \$3000, then the social cost of corruption is \$80,000. In other words, social costs of misdirection of resources toward activities that offer better corruption opportunities can vastly exceed bribe revenues.

Western observers often wonder about the preference for unnecessarily advanced rather than "appropriate" technology by Third World governments. Overinvoicing provides the obvious explanation for this preference for advanced technology. The rational managers and bureaucrats in poor countries want to import goods on which bribes are the easiest to take, not the goods that are most profitable for the state firms. To do that, they basically discourage or even prohibit the importation of appropriate technology, and encourage the importation of unique goods on which overpayment and overinvoicing are more difficult to detect. As a result, very poor countries end up with equipment way beyond their needs.

This example fits neatly into our framework. To maximize the value of their personal revenues, bureaucrats prohibit imports of goods on which bribes cannot be collected without detection, and encourage imports of goods on which they can collect bribes. As a consequence, the menu of both consumer and producer goods available in the country is determined by corruption opportunities rather than tastes or technological needs. This argument might suggest why so many poor countries would rather spend their limited resources on infrastructure projects and defense, where corruption opportunities are abundant, than on education and health, where they are much more limited. In light of the enormous returns on these forgone health and education projects, the social costs of corruption might be enormous. Without the need to keep corruption secret, officials could collect their bounty in much less distortionary ways.

The imperative of secrecy entails another potentially important cost of corruption, namely its hostility to change and innovation. Keeping corruption secret requires keeping down the number of people involved in giving and receiving bribes. The elite must then include only a small oligarchy of politicians and businessmen, and refuse entry to newcomers. This situation may well describe the Philippines under Marcos, Russia under Communists, or some African dictatorships. But innovation and change are often precipitated by outsiders. To the extent that the elite prevents them from entering, to maintain their profits or simply to keep down its numbers to preserve secrecy, growth will suffer. It remains an interesting puzzle how small ruling elites in Korea have managed to keep up innovation and growth despite the effective exclusion of outsiders from both economic and political participation.

V. Conclusion

This paper has explored two broad reasons why corruption may be costly to economic development. The first reason is the weakness of central government, which allows various governmental agencies and bureaucracies to impose independent bribes on private agents seeking complementary permits from these agencies. When the entry of these agencies into regulation is free, they will drive the cumulative bribe burden on private agents to infinity. A good illustration of this problem is foreign investment in post-Communist Russia. To invest in a Russian company, a foreigner must bribe every agency involved in foreign investment, including the foreign investment office, the relevant industrial ministry, the finance ministry, the executive branch of the local government, the legislative branch, the central bank, the state property bureau, and so on. The obvious result is that foreigners do not invest in Russia. Such competing bureaucracies, each of which can stop a project from proceeding, hamper investment and growth around the world, but especially in countries with weak governments.

Downs [1967] calls the expansion of bureaucracies into new regulations "territoriality," but does not elaborate on its consequences for resource allocation. We showed how costly territoriality can be when different agencies are neither kept honest nor controlled by a central authority. We have explored the effects of territoriality when agencies impose regulations independently to maximize their individual bribe revenues. But even if bureaucrats are kept honest and introduce regulations only to expand their own domains without coordination from above, compliance with these regulations can be very costly to private agents.

The second broad reason that corruption is costly is the distortions entailed by the necessary secrecy of corruption. The demands of secrecy can shift a country's investments away from the highest value projects, such as health and education, into potentially useless projects, such as defense and infrastructure, if the latter offer better opportunities for secret corruption. The demands of secrecy can also cause leaders of a country to maintain monopolies, to prevent entry, and to discourage innovation by outsiders if expanding the ranks of the elite can expose existing corruption practices. Such distortions from corruption can discourage useful investment and growth.

Throughout the paper we have argued that economic and political competition can reduce the level of corruption and its adverse effects. If different agencies compete in the provision of the same services, corruption will be driven down provided that agents cannot simply steal. Similarly, political competition opens up the government, reduces secrecy, and so can reduce corruption provided that decentralization of power does not lead to agency fiefdom and anarchy.

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REFERENCES

Banfield, Edward, "Corruption as a Feature of Government Organization," *Journal of Law and Economics*, XVIII (1975), 587–605.
Barro, Robert J., "Economic Growth in a Cross Section of Countries," *Quarterly Journal of Economics*, CVI (1991), 407–44.
Becker, Gary S., and George J. Stigler, "Law Enforcement, Malfeasance, and the Compensation of Enforcers," *Journal of Legal Studies*, III, (1974), 1–19.
Business International Corporation, *Introduction to the Country Assessment Service* (New York: Business International Corporation, 1984).

Carino, Ledivina V., ed., *Bureaucratic Corruption in Asia: Causes, Consequences, and Controls* (Quezon City, The Philippines: JMC Press, 1986).
Demsetz, Harold, "Why Regulate Utilities?" *Journal of Law and Economics,* XI (1968), 55–66.
De Soto, Hernando, *The Other Path* (New York: Harper and Row, 1989).
Downs, Anthony, *Inside Bureaucracy* (Boston, MA: Little, Brown, 1967).
Ekelund, Robert B., and Robert D. Tollison, *Mercantilism as a Rent Seeking Society* (College Station, TX: Texas A&M University Press, 1981).
Ekpo, Monday, *Bureaucratic Corruption in Sub-Saharan Africa: Toward a Search of Causes and Consequences* (Washington, DC: University Press of America, 1979).
Gould, David J., and Jośe A. Amaro-Reyes, *The Effects of Corruption on Administrative Performance,* World Bank Staff Working Paper No. 580 (Washington, DC: The World Bank, 1983).
Heilbroner, Robert, *The Making of Economic Society* (Englewood Cliffs, NJ: Prentice-Hall, 1962).
Huntington, Samuel P., *Political Order in Changing Societies* (New Haven, CT: Yale University Press, 1968).
Klitgaard, Robert, *Controlling Corruption* (Berkeley, CA: University of California Press, 1988).
____, *Tropical Gangsters* (New York: Basic Books, 1990).
____, "Gifts and Bribes," in Richard Zeckhauser, ed., *Strategy and Choice* (Cambridge, MA: MIT Press, 1991).
Krueger, Anne P., "The Political Economy of a Rent-Seeking Society," *American Economic Review,* LXIV (1974), 291–303.
Leff, Nathaniel, "Economic Development through Bureaucratic Corruption," *American Behavioral Scientist* (1964), 8–14.
Mauro, Paolo, "Country Risk and Growth," mimeo, 1993.
Rose-Ackerman, Susan, "The Economics of Corruption," *Journal of Public Economics,* IV (1975), 187–203.
____, *Corruption: A Study of Political Economy* (New York: Academic Press, 1978).
Shleifer, Andrei, and Robert W. Vishny, "Pervasive Shortages under Socialism," *Rand Journal of Economics,* XXIII (1992), 237–46.
Stigler, George J., "A Theory of Oligopoly," *Journal of Political Economy,* LXXII (1964), 44–61.
Tullock, Gordon, "The Welfare Cost of Tariffs, Monopoly and Theft," *Economic Inquiry,* V (1967), 224–32.
United Nations, *Corruption in Government* (New York: United Nations, 1989).